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A V2V Charging Allocation Protocol for Electric Vehicles in VANET

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Abstract

This paper tackles the problem of the propagation of charging information in Vehicular Ad-hoc Networks (VANETs) and the decentralized allocation of Vehicle-to-Vehicle (V2V) charging pairs. Electric Vehicles (EVs) have gained the attention of various sectors as a green alternative to the air-polluting fuel-based vehicles. However, due to the limitation of the EVs' battery capacity and the shortage in the number of charging stations causing significant waiting time to charge, V2V charging has emerged as a promising solution to help address these issues. Mainly, V2V solutions need to define two mechanisms: an information dissemination mechanism and a charging pairs allocation mechanism. In the literature, two architectures are proposed: centralized and semi-distributed using VANET. However, in addition to suffering from the single point of failure problem, centralized solutions consider only connected vehicles during the charging pair allocation. On the other hand, existing VANET based solutions address the problem of charging information dissemination while still performing the allocation of V2V charging pairs in a centralized way. In this paper, we propose an ad-hoc-based solution for V2V charging where both information dissemination and charging pairs allocation are performed over VANET. To do so, at the application layer of VANET, a decentralized protocol for V2V is proposed. This protocol is divided into two phases: *providers announcement* where consumers in the network can discover the nearby providers, and *providers-consumers allocation* where providers select the appropriate consumer based on the received $OFFER$s (Table 5). At the network layer, Quality of Service-Optimized Link State Routing (QoS-OLSR) is used to find routing paths using the Multi-Point Relays (MPRs). These MPRs are in fact utilized by the protocol to propagate the announcement of the providers to the network. The simulation results show high performance of the proposed protocol in terms of packet delivery ratio and end-to-end delay. In addition, in terms of payoff and allocation rate, the proposed protocol performs better than centralized allocation in the presence of disconnected EVs.

Keywords: Electric Vehicle, V2V Charging, QoS-OLSR, protocol, internet of vehicles, Ad-hoc

1. Introduction

Recently, Electric Vehicles (EVs), as part of Intelligent Transportation System (ITS), have become popular and gained the attention of car owners, manufacturers, and the support of leading countries, such as China, USA, Canada, and Japan [1, 2]. They have emerged as a green solution to the environmental issues caused by oil-driven cars and their polluting gas emissions. The European Union has set a goal to decrease the vehicles with CO2 emissions by 42% [2]. The demand on owning EVs is increasing and the worldwide EVs sales have increased from 0.55 million EVs in 2015 to 2.2 million EVs by the end of 2019 [3]. However, the main challenge that is facing the EVs industry is the battery lifetime [4]. Therefore, an adequate number of Charging Stations (CSs) is necessary for EVs to reach their destinations, especially during long-distance trips.

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Although car manufacturers keep enhancing the EV battery specifications to make it charge for longer durations, EV drivers still need to frequently charge their vehicles especially during long-distance trips for example on highways [4]. It is estimated that in the near future by 2030, the number of EVs owners will exceed 250 million [5]. As the number of EVs is increasing rapidly, the increase in the number of CSs are comparatively lower during recent years leading to an insufficient number of CSs [6, 7]. This puts a high load on the existing infrastructure, especially during peak hours. Additionally, the long charging time of the EVs leads to crowded charging stations and long queuing time [1, 6]. Considering the energy storage abilities of EVs and their capability to discharge and share excess energy with other EVs through bidirectional chargers, Vehicle-to-Vehicle (V2V) charging technology is fast emerging as a practical and efficient solution to reduce the load on the existing charging stations [8, 9, 7]. There are two main challenges in implementing V2V charging: 1) managing the allocation of V2V charging pairs, namely consumers (need charge) and providers (having additional charge), and 2) disseminating the charging information.

To collect charging information required to allocate V2V charging pairs, centralized solutions have been proposed in several works [1, 10, 11, 12]. However, central solutions suffer from multiple disadvantages such that it is - 1) infrastructure-dependent: The communication with central server requires full network coverage to discover and allocate EVs, that is mostly not practical in low coverage areas such as rural areas and areas with frequent disconnections such as highways. In addition, the infrastructure-based networks that are considered costly [13]. Also, central solutions are considered a 2) single point of failure: a breakdown in the central server will disconnect it from participating EVs causing interruptions in the allocation service [14]. To solve the aforementioned challenges of infrastructure-based centralized solutions, Vehicular Ad-hoc Networks (VANETs) are used as a distributed solution to communicate charging information for the allocation of V2V charging pairs. VANETs are self-organized and scalable networks that allow the propagation of information utilizing vehicles to everything (V2X) communication links [14]. Although VANET is highly dynamic, it is characterized by a moderate communication performance depending on the routing protocol used. Optimized Link State Routing (OLSR) was proved to be most suitable for VANET [15]. Therefore, various variants of OLSR protocol have been developed as Quality of Service-OLSR (QoS-OLSR) to enhance the performance in various environments, namely, MANET, urban-VANET, and highway-VANET [16, 17, 18]. Multiple works utilized VANET and its routing protocols for data delivery of V2V charging, such as the works proposed in [14, 13]. VANET was also utilized for propagating information to schedule slots for EVs at CSs for the purpose of managing the charging load at various CSs, and minimizing the queuing time [2, 19, 20, 21]. In the current state of the art, many techniques and algorithms are being used to address the problem of V2V charging pairs or CS selection and scheduling, including behavioral game theory, fog computing, and matching theory. Although VANET was proposed for information sharing, the existing solutions still rely on central unit to gather the information of the consumers and producers in the area of interest and perform the charging pair allocation task. To the best of our knowledge, none of the related works have utilized VANET in a fully-decentralized allocation of EVs for V2V charging.

This paper proposes a fully-decentralized V2V charging application layer protocol for VANET to allocate V2V charging pairs, namely consumers and providers. The *consumers* are the EVs with batteries' level below a threshold and are in need of charging to complete the trip, while *providers* are EVs that possess extra battery charge and are willing to sell it to other EVs. In addition to the V2V charging pair allocation task, the proposed protocol allows providers to announce themselves to all reachable EVs through 4-hop relays, known as Multi-Point Relays (MPRs). At the network layer, QoS-OLSR [16] is utilized to discover routing paths and build routing tables through the special nodes, MPRs. Although various QoS metrics have been proposed to improve OLSR performance, they were mostly designed for MANET [18] or for VANET in urban environment [22, 23]. However, the QoS proposed in [16] includes metrics suitable for highway-VANET, namely, velocity and traveled distance metrics. As the V2V charging allocation protocol is proposed to solve the issues of charging over long distances and provide connectivity for vehicles in such environments, for example at highways and rural areas, the QoS-OLSR proposed in [16] is implemented as an underlying protocol to connect vehicles at the network layer. In QoS-OLSR [16], MPRs are responsible to route packets for their selectors. The proposed protocol also utilizes the MPRs at the application layer to announce the providers to the network. A set of messages is proposed to fulfill the tasks of the proposed protocol, which are charging pairs allocation and providers announcement. Every MPR sends a *Providers Announcement*

(PA) message, that contains the set of MPR selectors who are providers, to 4-hop vehicles. The PA message holds the provider's IP address, energy selling price, and availability time, for every provider added to the message. MPRs also participate in propagating other MPRs' PA messages by forwarding them once they are received. Based on the received PA messages, each consumer proposes to the provider with minimal payment by sending an $OFFER$ message specifying the required energy and preferred meeting point. Providers accept the consumers' $OFFER$ with the maximum payoff, by sending an $ACCEPT$ message specifying the meeting point that is suitable for both parties, the consumer and the provider.

Hence, the main contribution of this paper is an ad-hoc V2V charging protocol that:

- Provides reachability and connectivity for providers and consumers in infrastructure-less settings.
- Allows the ad-hoc discovery of providers by designing and implementing new protocol exchanged messages.
- Allocates provider-consumer pairs for V2V charging in fully decentralized and ad-hoc manner with reasonable delay.

Simulations are conducted using Network Simulator 3 (NS3) and Sumo to compare the proposed protocol to the centralized solution in the presence of disconnected EVs. In terms of payoff and allocation rate, the proposed protocol performs as the full connected central allocation with minimal difference. While the ad-hoc allocation protocol performs better than central allocation with limited connectivity.

The rest of the paper is organized as follows. In Section 2, related works and their limitations are discussed. Following Section 3, where the proposed protocol is explained in detail. The protocol messages are presented in Section 4, followed by an illustrative example in Section 5. The simulation environment, parameters, and results are detailed in Section 6. Finally, Section 7 presents the conclusion of this work.

2. Related Work

In the literature, many solutions were proposed to address the problem of charging pair allocation and data dissemination. These solutions can be categorized to: centralized and semi-distributed.

2.1. Centralized Architectures

Various works have proposed centralized architectures to tackle the issue of V2V charging and charging slot scheduling at CSs, using different algorithms and models. To address the *V2V charging* problem, the authors of [10] proposed a central mechanism for the allocation of charging pairs. The EVs' owners that require charging send first a request to the central unit. The main challenge targeted by [10] is the distribution of suppliers and demanders such that the supplier is available and close enough to the demander by a distance less than the demander's remaining range. Therefore, two assignment models are proposed based on the system and individuals' benefits. Also, in [11], the authors proposed a central solution for the energy trading assignment model that is hosted on the cloud. Charging and discharging EVs submit bids and aspiration levels to the cloud that matches the charging pairs based on a behavioral energy trading game. The proposed model does not consider the preferences of EVs to ensure privacy and auto-compute the aspiration levels to ensure fairness. Also, the authors of [6] tackle the problem of EVs rage anxiety and propose a matching game theory approach to allocate energy consumers and providers for V2V charging by considering their preferences which lead to an increase in the satisfaction of EVs' owners. Additionally, a V2V charging energy management framework is proposed in [7] that maximizes allocation of charging pairs, achieves satisfaction of users, and is effective in terms of cost. The framework considers various parameters including, trading prices, EVs charge state, in addition to the users' rationality. The stages of the V2V charging framework are introduced and possible solutions are discussed at each stage. The introduced stages are, the discovery of EVs and data gathering, making decisions for V2V pair allocation, V2V energy transfer, and privacy of information and security of payments. In [12], the authors proposed a model based on fog computing to support energy trading among hybrid electric vehicles. The main goal of the work is to locally manage the energy trading by local fog computing centers and maximize vehicles profits by minimizing

time cost and energy loss. Two models are proposed, non-profit and profit fog computing centers, that are designed as a single-objective interior point method and multi-objective genetic algorithm respectively. The fog computer centers are distributed such that each one covers a certain area and communicates with vehicles through wireless communication. These fog centers are responsible for executing the pricing and scheduling algorithms. Authors of [24] also proposed routing and scheduling algorithms for V2V charging where EVs are scheduled to charge from charging trucks. The main goal of the work [24] is to find the optimal routes of the charging trucks that maximize the number of EVs charged.

Other works tackled the problem of *scheduling EVs at charging stations*. In [21], authors tackle the issue of selecting charging stations and routing EVs aiming to minimize the total charging time. Through the means of crowd-sensing, charging-related information, such as the battery, traffic, and queues statuses, are utilized to schedule EVs optimally. Minimize charging cost has been tackled in [25] however for electric buses. The problem in [25] is presented as a non-linear program then reformulated as a linear program and solved accordingly. Bids were considered in [26] to schedule charging slots to EVs. The authors in [26] proposed a scheduling algorithm namely rolling horizon by the means of a genetic algorithm aiming to maximize energy distributors' profit and minimize EVs cost. In [27], the authors proposed a reservation management model for EVs at CSs to avoid congestion. At CSs, the EVs switch their batteries with fully charged batteries. The main goal of the reservation system is to decrease the waiting time of EVs. Therefore, the selection of CS for an EV that requested charge is based on the CS with a maximum number of charged batteries. In case none of the CSs has a charged battery, the CS is selected based on the lowest waiting time.

2.2. Semi-Distributed Architectures

While fully-centralized architectures suffer from the limitations of being required to be connected to all EVs for best performance, various works introduced data delivery through VANET to allocate charging pairs. Tackling the problem of *V2V charging*, the authors of [13] proposed a semi-centralized framework for managing charging navigation for electric vehicles, divided into local and global schemes. They utilized VANET for communication, Mobile Edge Computers (MECs) communicate stopping locations status and decisions between EVs and navigation control center. The distributed local navigation scheme, carried out by every EV, results in a decision of the optimal traveling route and subsequently the most available stopping location. These decisions are based on traveling time prediction, charging time, and charging comfortable degree models proposed by the authors. After the local decisions are shared with the control center through MECs, it executes a global charging navigation schemes that result in charging-discharging matching pairs. The matching problem is solved through Kuhn-Munkres-based optimal matching algorithm. The work is extended in [14] to tackle the problem of charging data delivery through the means of reinforcement learning. The work proposed in [28] also introduces VANET for the propagation of data between EVs and a parking service center for the purpose of V2V charging. The proposed V2V charging scheme in [28] works on selecting a parking slot and matching charging pairs. The parking slot reservation is based on minimizing travel cost and maximizing charging pleasure (defined as the waiting time between arriving at the parking slot and begin charging). While a first-come, first-serve algorithm is implemented for matching EVs based on the earliest arrival time. Also, the work in [1] tackles the problem of matching charging EVs pairs for V2V charging. The authors proposed a max-weight matching algorithm that is further enhanced by a stable matching algorithm that considers the rationality of EVs. The optimality of the matching algorithm is either consumer-oriented or provider-oriented. The charging-discharging pairs matching algorithm is executed at a central server that collects real-time information of EVs, parking lots, and other power-related entities. The EVs are assumed to be part of the ITS that communicates with roadside units to share their information.

In [29], a CS reservation model is proposed where the decision is made by EVs. The work proposes a communication model based on VANET between EVs, edge servers, and CSs. The CSs proactively update their status at the edge servers, while edge servers update the EVs with the CSs status in a reactive manner at the request of the EV. Ant-colony optimization is used to solve the issue of selecting the optimal data routing path which is defined as the route with maximum QoS. The QoS is calculated based on the transmission delay and delay limit of the route. Finally, a waiting time model is formulated which contributes to improving the decision of local relaying.

Addressing the issue of *charging station slots scheduling and allocation*, the authors of [2] proposed a charging stations selection scheme for EVs. The main goal of the work is to minimize the overall trip time of EVs towards the selected charging station. The selection of charging station-EV pair is performed by a central management server. The work utilizes VANET for the communication between EVs and Road Side Units (RSUs). For an EV to get allocated to a charging station, it sends a request message, proposed by the authors, to the nearest RSU. Then, the RSU forwards the request to the central server which in return replies back with the most suitable allocated charging station. All messages are communicated through V2X VANET communication by the means of IEEE 1609 WAVE technology. The allocation solution is centralized. In addition, VANET is utilized only between vehicles and RSUs such that the work lacks vehicle-to-vehicle communication. A central management unit namely Global Aggregator (GA) is employed in [19] for allocating charging stations and scheduling charging slots for the EVs. An EV that needs a charge sends a request directly to the GA or through V2V links of VANET. Hence, EVs use VANET to relay messages toward the GA and vice versa. The GA then replies with allocated charging stations. Once the EV reaches the destination charging station, it waits for the scheduling of the charging slot by the GA. Additionally, an architecture for assigning charging slots for EVs has been also proposed in [20] utilizing EVs to everything communication models. The main objective of the work [20] is to minimize charging time, cost, travel distance, and queuing delay. In [30], the authors tackle the problem of advertising charging stations to surrounding electric vehicles. Therefore, they propose an incentive model based on a reward scheme to motivate Plug-in Electric Vehicles (PEVs) to advertise the charging stations. Vehicles and charging stations communicate through VANET V2V and V2I communication links. Hence, VANET is utilized in [30] to relay information packets. Two reward policy mechanisms are proposed; pre-determined and optimal policies utilizing fairness task time allocation and Stackelberg game respectively. A solution is also proposed by the authors of [31] based on Mobile Edge Computers (MECs) for CSs Selection problem. EVs, MECs, and a cloud server communicate through VANET links, V2V, Vehicle-to-Edge (V2E), and Vehicle-to-Cloud (V2C). The MECs broadcast the corresponding CSs earliest charging time information to the EVs through the cellular network. Based on the received information, each EV selects a CS. The reservation is relayed to the cloud or to a MEC. The cloud server handles the EVs that arrive at the charging station later than a pre-determined time limit. Finally, the charging station updates the clouds with the list of reservations and queue list where the cloud uses them to calculate the new earliest available charging time. VANET in this work is only utilized to relay information.

The works proposed in [13, 31, 12] are semi-distributed solutions for the allocation of V2V charging pairs while in [2, 19, 10, 11, 1], the authors proposed a fully centralized solutions. VANET is utilized in the related works for the means of relaying data between central unit, charging stations, and vehicles. As the allocation performance of central servers or cloud is limited by the connectivity with EVs, VANET provides better reachability among vehicles and considers their rationality. Therefore, the proposed protocol in this paper is implemented in an ad-hoc network to disseminate information and allocate V2V charging pairs.

2.3. Network Layer Protocols for VANET

To disseminate charging data over VANET, a suitable routing protocol should be adopted to provide adequate connectivity. Among the various ad-hoc routing protocols, the reactive, proactive, and geographical-based protocols, the work in [15] has proved that the proactive protocol OLSR has the highest network performance in VANET. Consecutively, variants of OLSR have been proposed for MANETs, urban-VANETs, and highway-VANETs. A QoS-OLSR was implemented in [18] to improve the performance of MANET. QoS functions were also proposed in [32] and [33] to incorporate urban environment metrics such as traffic lights, number of lanes, and intersections to select MPRs of OLSR. Other routing protocols were proposed in various works to enhance communication performance in VANET such as in [34] where vehicles route packets between intersections and based on the traffic light statuses. Clustering techniques were also proposed for VANET and MANET by utilizing Chain-Branch-Leaf technique [35], Gale-Shapley matching game [36, 22], and evolutionary game theory[37].

The main goal of this paper is to provide reachability and connectivity over VANET for EVs to allocate them for V2V charging in the absence of connection with communication infrastructure. Highways and rural areas are the most common areas for frequent disconnections with infrastructure-based networks. The

work in [16] proposes a QoS-OLSR protocol for highway environments where it utilizes bandwidth, neighbors, traveling velocity, and distance as a QoS metric for MPRs selection. Therefore, it provides adequate network performance for the proposed application layer protocol, namely, the V2V charging protocol.

3. Proposed Protocol

3.1. Overview

A decentralized application layer protocol for VANET is proposed to discover nearby providers and allocate consumer-provider charging pairs for V2V charging. The network consists of a set of vehicles $V = \{v_1, ..., v_n\}$. In addition, the network includes a set of providers and consumers presented as $P = \{p_1, ..., p_m\}$ and $C = \{c_1, ..., c_k\}$ respectively such that $P \subseteq V$ and $C \subseteq V$. The vehicles communicate through V2V links of VANET utilizing IEEE 802.11p short-radio technologies. Through V2V links, vehicles communicate via the messages presented in Table 1 and further elaborated in Section 4. Each $v \in V$ is equipped with a Global Positioning System (GPS) and a map to determine its position and meeting points.

Table 1: Protocol Messages

Message	**Use**
$HELLO$ QoS-OLSR (Table 2)	Providers share their intention to provide energy with their neighbors
Providers Announcement (PA) (Table 4)	Propagated in the network to announce the providers' identities and their parameters such as selling price and availability time
$OFFER$ (Table 5)	Sent by a consumer to a provider requesting for allocation
$ACCEPT$ (Table 6)	Sent by a provider to a consumer to accept the charging $OFFER$
ACK	Sent by a consumer to acknowledge allocation
$ACKACK$	Sent by a provider to acknowledge allocation

At the network layer, the proposed protocol utilizes QoS-OLSR protocol [16]. In QoS-OLSR, every $v_i \in V$ is assigned a QoS value used for the selection of MPRs. The selected MPRs relay data and generate routing tables for their selectors. The QoS value is calculated based on the vehicle's velocity, position, number of neighbors and, bandwidth as presented in equation 1.

$$QoS_{v_i} = BW_{v_i} \times N_{v_i} \times \frac{D_{v_i}}{\nu_{v_i}} \tag{1}$$

where QoS_{v_i} is the QoS value of vehicle i, BW_{v_i} is the bandwidth, and N_{v_i} is the number of neighbors of vehicle i. D_{v_i} and ν_{v_i} are the distance ratio and velocity ratio, respectively. D_{v_i} is the ratio of remaining traveling distance over total distance, while ν_{v_i} is the ratio of the current velocity of the vehicle v_i to its average velocity [16].

3.2. V2V charging Pair Allocation Protocol

The consumer-provider allocation is proposed as a decentralized protocol for VANET. The first phase of allocation is the announcement of providers to the network. In this phase, the providers' MPRs are responsible for propagating their information to all reachable vehicles through 4-hop MPRs. After the consumers construct the set of nearby providers, the second phase begins. The consumers begin the allocation phase by sending requests to the providers. Consequently, providers reply to the most suitable requester. Finally, the selected consumer acknowledges the selected provider which replies with $ACKACK$ message. The provider and consumer are considered allocated once they receive the acknowledgments. Figure 1 shows the sequence diagram of the protocol highlighting the message exchange between vehicles.

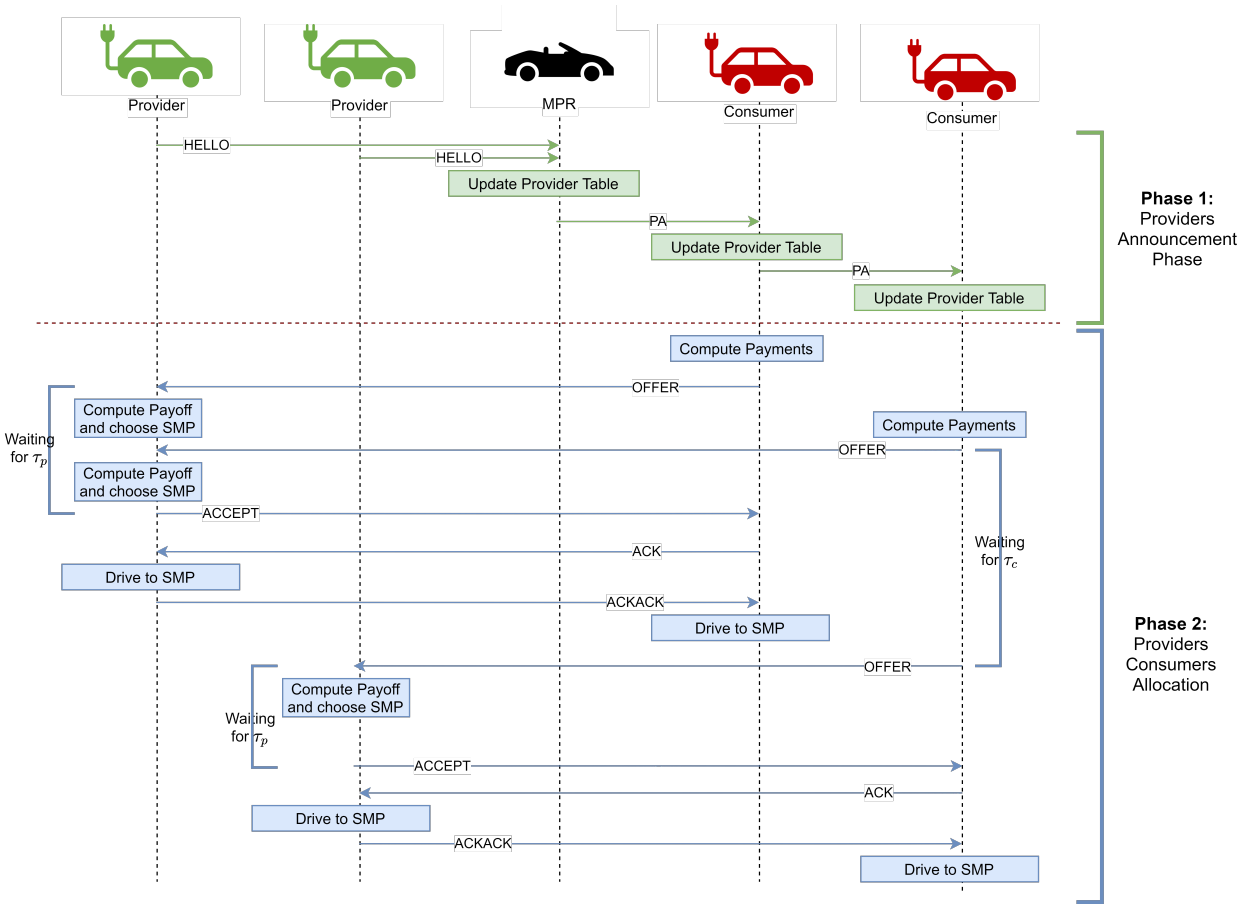


Figure 1: V2V charging Protocol Sequence Diagram

3.2.1. Providers Announcement Phase

The Providers Announcement Phase is essential to propagate the information of the providers to the network and for consumers to locate nearby providers. In this phase, the providers use the modified QoS-OLSR $HELLO$ message (Table 2) to share their information with the 1-hop neighbors. Such providers' information includes the selling price of energy unit and their availability duration at the current street. The MPR of each $p \in P$ receives the $HELLO$ message (Table 2) and adds or updates its Providers' Table. Every time duration τ_{pa}, the MPRs construct Providers Announcement (PA) message (Table 4) by adding the information of the providers that are MPR selectors. They also initialize and set the time to live (ttl) value of PA message (Table 4) to n. In this paper, $n = 4$, which means the message reaches 4-hop away MPRs. Simultaneously, an MPR forwards PA messages (Table 4) only if its time to live is larger than 0 and the sender is in its MPR Selector set. Once every $v \in V$ receives a PA message (Table 4), it processes it by updating its Providers Table. Every provider added to the Providers Table is considered a reachable provider to the consumer. Algorithm 1 presents the Providers Announcement algorithm. In the sequence diagram, presented in Figure 1, the *Provider Announcement Algorithm* is reflected in Phase 1 where providers shares $HELLO$ message (Table 2) with neighbor MPR. Following, the MPR begins the propagation of PA message (Table 4).

3.2.2. Providers-Consumers Allocation

After the providers' information is shared with the network, every consumer $c \in C$ constructs its Providers Table including the set of reachable providers with their energy unit price and availability time

at the street segment. The consumers begin the allocation process by calculating a payment value for each reachable provider in the Providers Table. The payment value for each reachable provider is calculated according to equation 2.

Algorithm 1 Providers Announcement Phase

P = Set of providers in the network
$N(p)$ = Set of 1-hop neighbors of provider p
$price(p)$ = The selling energy unit price of provider p
$availability(p)$ = The time provider p is available to charge computed in *seconds*
MPR = The set of MPRs in the network
$MPRSelectors(mpr)$ = The set of MPR Selectors of MPR mpr
$P(v)$ = The set of providers in the 1-hop neighbors of vehicle v
$ProvidersTable(v)$ = The Providers Table of vehicle v
ttl = Time to live of PA message

for Each provider $p \in P$ **do**
    Add $price(p)$ and $availability(p)$ to QoS-OLSR $HELLO$ message
    Share $HELLO$ message with $N(p)$

for Each $mpr \in MPR$ **do**
    Add the set of providers $P(mpr) : P(mpr) \subseteq MPRSelectors(mpr)$ to the PA message
    $ttl = n$
    Broadcast PA message to $N(mpr)$
    if received PA message from vehicle $v \in MPRSelectors(mpr)$ AND $ttl > 0$ **then**
        Forward received PA message
        $ttl = ttl - 1$

for Each vehicle v in the network **do**
    if received PA message **then**
        Update $ProvidersTable(v)$

$$Payment_p = UP_p \times RE_c \tag{2}$$

where UP_p is the selling price for energy unit of provider p and RE_c is the required energy of consumer c.

Then, each consumer $c \in C$ selects the reachable provider (p_s) with the least payment and sends it an $OFFER$ message (Table 5). Before constructing the $OFFER$ message (Table 5), each consumer c locates the nearest meeting point (PMP_c) according to his position. The PMP_c and RE_c are added to the $OFFER$ message (Table 5) and sent to the p_s. Once the $OFFER$ (Table 5) is sent, consumer c starts a timer waiting for the recipient of the $ACCEPT$ (Table 6) message. The steps of the providers selection and $OFFER$ message (Table 5) construction are presented in Algorithm 2, lines 1 to 8.

Referring to Algorithm 3, the providers begin the charging pair allocation process by receiving an $OFFER$ message (Table 5) for the first time. For the provider to select a consumer, it keeps track of all the received $OFFER$ messages (Table 5) within τ_o duration starting from the time the first $OFFER$ (Table 5) is received. Therefore, once an $OFFER$ (Table 5) is received within τ_o the provider adds it to the set of Offers and computes a payoff value for the corresponding consumer. The payoff is calculated according to equation 3.

$$Payoff_p = Payment_c - Cost_p \tag{3}$$

The total payoff of provider p is the net payment received from consumer c, where net payment is the

payment received from consumer c minus the total cost for providing a charge. The total cost ($Cost_p$) is computed in equation 4.

$$Cost_p = Cost_T + Cost_W + Cost_B + Cost_{Ch} \tag{4}$$

where $Cost_T$ is the Travel Cost of provider p to the selected meeting point (SMP) and $Cost_W$ is the Waiting time Cost. While $Cost_B$ is the Battery degradation Cost, the charging time Cost is presented as $Cost_{Ch}$.

$$Cost_T = Dis(loc_p, SMP) \times \alpha_p \times pr_o \tag{5}$$

$Cost_T$ as computed in equation 5 presents the price of traveling to SMP, such that, SMP is selected based on the common meeting point both, provider p and consumer c, can meet at. Thus, if PMP_p is farthest than PMP_c then SMP is PMP_p and vice versa. To compute $Cost_T$, the provider calculates the distance between its location and SMP, ($Dis(loc_p, SMP)$, its consumption rate (α_p), and original trading price (pr_o).

$$Cost_W = \begin{cases} (t^c_{travel} - t^p_{travel}) \times \zeta, & \text{if } loc_p > eloc_c \\ 0, & \text{if } loc_p < eloc_c \end{cases} \tag{6}$$

$Cost_W$ is shown in equation 6 and presents the cost of waiting time of the provider for the consumer to reach SMP. ζ in equation 6 is the time value computed in dollars/hour. In case the provider's location (loc_p) is a head of the consumer's estimated location ($eloc_c$), then $Cost_W$ is the travel time (t_{travel}) difference of consumer and provider. However, if the opposite applies, then the waiting time of the provider is 0. Subsequently, the waiting cost is 0. t_{travel} of the consumer/provider is computed according to equation 7.

$$t^v_{travel} = \frac{Dis(loc_v, SMP)}{sp_v} \tag{7}$$

where, t^v_{travel} is the travel time of vehicle $v \in V$ to SMP and sp_v is the speed of v.

Travel times t^c_{travel} and t^p_{travel} are both computed by the provider at the time it receives the $OFFER$ (Table 5) from consumer c. To calculate t^c_{travel}, the provider estimates the location of c based on PMP_c. As the consumer c sends the nearest meeting point in the $OFFER$ message (Table 5), the provider determines c's position according to equation 8 and its speed as half the street's maximum speed.

$$eloc_c = \frac{loc(PMP_c) - loc(PMP_c - 1)}{2} \tag{8}$$

where $loc(PMP_c)$ is the position of PMP_c and $loc(PMP_c - 1)$ is the position of the meeting point precedent to PMP_c.

$Cost_B$ also contributes to the total cost calculation. As in equation 9, $Cost_B$ depends on the battery replacement cost (β), degradation coefficient (ψ) and the energy amount to be transferred.

$$Cost_B = \beta \times \psi \times RE_c \tag{9}$$

Finally, equation 10 presents the calculation of $Cost_{Ch}$.

$$Cost_{Ch} = \frac{RE_c}{\eta \times \omega} \times \zeta \tag{10}$$

where η is the charging efficiency coefficient and ω is the charging rate.

Once the receiving offers timer, τ_o, is up, the provider p selects the offered consumer (c_S) with maximum payoff as shown in Algorithm 3, line 10. Then, it sends $ACCEPT$ message (Table 6) to c_s containing the corresponding SMP.

At the c_s, the $ACCEPT$ message (Table 6) is received within τ time duration. Hence, it replies to p_s with ACK message acknowledging the allocation acceptance. However, if the a consumer did not receive

Algorithm 2 Consumers' Allocation Algorithm

C = Set of Consumers in the network
$PT(c)$ = Providers Table of Consumer c
$pt(p)$ = Tuple pt containing provider p
PMP_c = Preferred meeting point of consumer c
$Payment(p)$ = Payment for provider p

for Each provider tuple $pt(p) \in PT(c)$ **do**
    Compute $Payment(p)$ according to equation 2
while $size(PT(c))$ AND $ACCEPT$ not received **do**
    Choose nearest meeting point as PMP_c
    Construct $OFFER$ message with RE_c and PMP_c
    Send $OFFER$ to the selected provider p_s such that $Payment(p_s) = min(\forall Payment(p) \mid p \in PT(c))$
    Start timer τ
    while τ **do**
        if received $ACCEPT$ message from provider p_s **then**
            Send ACK message
            Wait to receive $ACKACK$ message
            if $ACKACK$ is received **then**
                c is Allocated
            $\tau = 0$
    if $ACCEPT$ not received **then**
        $PT(c) = PT(c) - pt(p_s)$

Algorithm 3 Providers' Allocation Algorithm

P = Set of Providers in the network
C = Set of Consumers in the network
$SMP(c)$ = Selected meeting point of consumer c
$OFFERS(p)$ = Set of Offers received by provider p
$Payoff(c)$ = Payoff of consumer c
τ = Timer for the providers to consider offers

if received $offer$ from consumer $c \in C$ **then**
    $OFFERS(p) = OFFERS(p) \cup offer$
    Select $SMP(c)$ as the nearest common meeting point
    Compute $Payoff(c)$ according to equation 3
    Start timer τ
for τ **do**
    if received $offer$ from consumer $c \in C$ **then**
        $OFFERS(p) = OFFERS(p) \cup offer$
        Compute $Payoff(c)$ according to equation 3
Select consumer c_s such that $Payoff(c_s) = max(\forall Payoff(c) \mid c \in OFFERS(p))$
Set SMP as $SMP(c_s)$
Send $ACCEPT$ message to c_s including SMP
if received ACK from c_s **then**
    p is Allocated
    Send $ACKACK$ to c_s

an $ACCEPT$ message (Table 6) within τ time, it sends an $OFFER$ (Table 5) to the next least payment

reachable provider. The later steps are shown in Algorithm 2, lines 10 to 16.

At p_s side, as shown in Algorithm 3, lines 13 to 15, it receives the ACK message from c_s and replies with $ACKACK$ to acknowledge the allocation. The recipient of ACK message means the provider is allocated to c_s.

Algorithms 3 and 2 are reflected in Figure 1 in Phase 2. In the sequence diagram, consumers compute payments and send $OFFER$ message (Table 5) within τ_p time duration. When τ_p time up the provider sends $ACCEPT$ message (Table 6) to the consumer based on payoff. Accepted consumer replies with ACK followed by $ACKACK$ from providers side. Eventually, both, provider and consumer, drive to SMP. The non-allocated consumers repeat the process with the next available providers.

4. Protocol Messages

In this section, the proposed protocol messages are explained in addition to the structures used to complement the consumer-provider charging pair allocation process. The messages include the $HELLO$ message (Table 2) of QoS-OLSR that is modified and broadcast at the network layer. However, all the new proposed messages are sent to a specific destination vehicle at the application layer, utilizing the routing table generated at the network layer.

The Modified QoS-OLSR $HELLO$ **message** is presented in Table 2. The $HELLO$ message (Table 2) is a network layer message that is broadcast to all 1-hop neighbors. The added fields, shown in bold in Table 2, are *QoS*, *Street*, *Provider*, *Selling Unit Price*, and *Availability* time. *QoS* and *Street* fields are used for the selection process of MPRs at the network layer. However, the latter three parameters are used by the providers to announce themselves to their MPRs. *Provider* parameter is a Boolean value that is 1 if the vehicle is a provider and 0 otherwise. *Selling Unit Price* and *Availability* are 16 bit unsigned integers representing the values for the corresponding provider. Non-providers vehicles fill these fields with 0 value. Vehicles that processes $HELLO$ message (Table 2) will ignore the latter two parameters if *Provider* field is 0. When a provider p broadcast $HELLO$ message (Table 2), its MPR processes it by adding the provider to its **Providers Table**.

Table 2: Extended $HELLO$ Packet Format

0		16	31
Reserved		Htime	Willingness
QoS		**Street**	
Provider		**Reserved**	
Selling Unit Price		**Availability**	
Link Code	Reserved	Link Message Size	
Neighbor Interface Address			
...			

The Providers Table structure is created at every $v \in V$, such that consumers and MPRs utilize it to determine reachable providers and the set of providers to announce to the network respectively. The Table columns consist of: Provider IP Address (IP), Selling Unit Price (cents/kWh), Availability Time (seconds), Tuple Hold Time (seconds), Message's Sequence Number, and Source IP Address (IP).

The latter two parameters are used to determine the IP of the provider and the most recent message in case more than one message has been received for the same provider. The tuples are added or updated by the processing of PA (Table 4) or $HELLO$ messages (Table 2).

The protocol **Messages' Header Format** is presented in Table 3, which is inspired by the original OLSR message format [38]. All vehicles use this format to create application layer messages explained in this section. The common message header is first added and then the specific message header is added to the *Message* field. *Message Type* is used to determine which application layer message the *Message* field contains. The *Message Size* is utilized jointly with the *Message Type* for vehicles to process the *Message* correctly. The *Message Type* allows vehicles to identify the message and process it accordingly, while the

Message Size main purpose is to read the *Message* from the buffer according to its size. The *Message Sequence number* is updated every time the sender sends a new message. However, *Time to Live* is updated when a vehicle forwards a message. Finally, *Source Address* allows the tracking of the original sender of the message.

Table 3: Messages' Header Format

0		16	31
Message Type	Reserved	Message Size	
Sequence Number		Time to Live	
Source Address			
Message			

The **Provider Announcement message (PA)**, shown in Table 4, is broadcast by providers' MPRs and processed by all vehicles in the network. MPRs that have providers in their *MPR Selectors* set, create the PA message (Table 4) by adding every provider's *IP Address*, *Selling Unit Price* and *Availability time*. PA (Table 4) propagates to n-hop nodes, beginning from the MPR. The propagation technique used for PA messages (Table 4) is the same as Topology Control messages of OLSR [38]. MPRs that receive a PA message (Table 4), forward it only if the sender is its MPR Selector. PA message (Table 4) main goal is for consumers to locate nearby providers, hence, they can begin the process of finding a suitable charge provider.

Table 4: Providers Announcement message Format

0	16	31
First Advertised Provider IP Address		
Selling Unit Price	Availability	
Second Advertised Provider IP Address		
Selling Unit Price	Availability	
...		

The $OFFER$ message is a protocol message that is initiated and sent by consumers. Each consumer begins the charging pair allocation process by sending $OFFER$ (Table 5) message to a selected provider. As shown in Table 5, the $OFFER$ (Table 5) message contains the *Required Energy* of the consumer and its nearest meeting point which is PMP. The providers consider the offers of the sender for the selection within a specific time frame. They add the $OFFER$ (Table 5) received to **Consumer Offer Table**.

Table 5: $OFFER$ message Format

0	16	31
Required Energy	Preferred Meeting Point	

The **Consumer Offer Table structure** is created and updated at each provider. The main parameters of the table are:

- Offered Consumer IP Address (IP)
- Required Energy (kWh)
- Consumer's Nearest Meeting Point
- Selected Meeting Point
- Payoff (cents)

Once the $OFFER$ (Table 5) is received within the time frame, the recipient provider adds a new tuple by adding the *Consumers IP Address*, *Required Energy*, and consumer's *Nearest Meeting Point* from the information provided in the $OFFER$ message (Table 5). Based in these information, the provider computes the *Payoff* and selects a suitable meeting point which is the *Selected Meeting Point*.

The $ACCEPT$ message, presented in Table 6, is sent by a provider to the consumer it selected. Since the provider determines the *Selected Meeting Point* based on the preference of the offered consumer, it adds this information to the $ACCEPT$ message (Table 6). The recipient consumer process the $ACCEPT$ (Table 6) by sending an Acknowledgment to the sender.

Table 6: $ACCEPT$ message Format

0	16 … 31
Selected meeting point	Reserved

ACK and $ACKACK$ **messages** are empty messages that are used to acknowledge the acceptance of the provider and the allocation respectively. A consumer sends ACK message to the accepted provider, thus the provider is considered allocated upon the receipt of the message. Consequently, the provider replies with $ACKACK$ message to the ACK message. Hence, the recipient consumer of $ACKACK$ is considered allocated to the acknowledged provider.

5. Illustrative Example

In this section, an illustrative example is presented to show the different phases of consumer-provider charging pairs allocation. Referring to the scenario presented in Figure 2, 7 electric vehicles traveling on a 3 lane highway, all in the same direction. The scenario also includes 2 meeting points for charging purposes. Each vehicle is characterized by a unique ID, as shown in the figure. As vehicles communicate through V2V links with neighbors, Table 7 shows the vehicles' connection links. At the network layer protocol, QoS-OLSR [16], the vehicles choose an MPR to manage the routing. Also, Table 7 shows the selected MPR of each vehicle. These MPRs are utilized in the proposed protocol for the providers' announcement to the network. In addition, considering the charging status of each EV as a charge provider, consumer, or neither, it is presented in Table 7. Vehicles v1 and v3 are the providers in the considered scenario, While vehicles v5 and v7 are the charge consumers. Tables 9 and 10 show the providers and consumers parameters respectively.

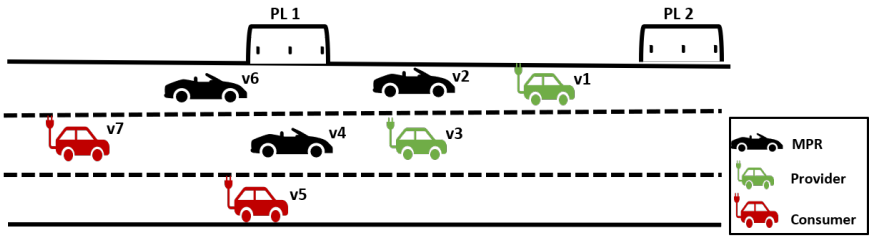


Figure 2: Illustrative Example Scenario

Table 7: Vehicles' Parameters

Vehicle ID	**v1**	**v2**	**v3**	**v4**	**v5**	**v6**	**v7**
Neighbors	{v2,v3}	{v1,v3,v4}	{v1,v2,v4}	{v2,v3,v5,v6}	{v4}	{v4,v7}	{v6}
MPR	v2	v4	v2	v6	v4	v4	v6
Charging Status	Provider	NA	Provider	NA	Consumer	NA	Consumer

Table 8: MPR Selectors Set

MPR	v2	v4	v6
MPR Selectors	{v1,v3}	{v2,v5,v6}	{v4,v7}

Table 9: Provider's Parameters

Provider	**Selling Energy Unit Price (cents/kWh)**	**Availability (seconds)**
v1	10	300
v3	13	300

Table 10: Consumers' Parameters

Consumer	**Required Energy (kWh)**	**Preferred Meeting Point**
v5	10	2
v7	15	1

At the network layer, providers share $HELLO$ message (Table 2) with the direct neighbors specifying their parameters, selling unit price, and availability time. The proposed application layer protocol begins with the propagation of the PA messages (Table 4). All MPRs are responsible for announcing the providers to the network. In this example, the MPR $v2$ is the only MPR that has providers, $v1$ and $v3$, in its *MPR Selectors* set as presented in Table 8. Therefore, it generates PA message (Table 4) and sends it to the network. The following illustrates the propagation of PA message (Table 4) generated by MPR 2.

Since vehicle $v2$ is the MPR of providers $v1$ and $v3$, it creates a PA message (Table 4) and broadcast it to the neighbors. MPR $v2$ appends the information of its MPR selectors that are providers as presented in Table 9. The PA message (Table 4) is broadcast and received by the vehicles, $v1$, $v3$ and $v4$. Recipient vehicles process the message by adding the information included in the PA message (Table 4) to its Providers Table. An example of the content of the Providers Table of vehicle $v4$ is shown in Table 11. Following, vehicle $v4$ is the MPR of $v2$, therefore, it forwards the received PA message (Table 4). Vehicles $v5$ and $v6$ receive the PA message (Table 4) and update their Providers Table. Vehicle v6 is the MPR of $v4$, hence, it forwards the PA message (Table 4) to reach vehicle $v7$. Finally $v7$ updates its Provider Table on the recipient of the PA message (Table 4) that is originated from vehicle $v2$. At this step, all connected vehicles in the scenario know that providers $v1$ and $v3$ exist and are reachable.

Table 11: Vehicle 4 Providers Table

Table Parameter	**Column 1**	**Column 2**
Provider IP Address	10.1.1.1	10.1.1.3
Selling Unit Price (cents\kWh)	10	13
Availability (Sec.)	300	300
Tuple Hold Time (Sec.)	300	300
Sequence Number	1	1
Source IP Address	10.1.1.2	10.1.1.2

Now, the consumers $v5$ and $v7$ needs to charge. Referring to their Providers Table, they compute the payment value for each provider according to equation 2. Table 12 shows the payment values for each provider computed by each consumer based on the values in Tables 9 and 10. Depending on these calculated payments, consumers $v5$ and $v7$ creates an $OFFER$ message (Table 5) with payment and preferred meeting point, then send it to provider $v1$, since provider $v1$ has the least payment for both consumers. Provider $v1$ receives the $OFFER$ messages (Table 5) within τ time duration. For each received $OFFER$ (Table 5), provider $v1$ adds it to the set of Offers. Once the timer is up, provider $v1$ computes the payoff according to equation 3. Table 13 shows the payoff for each consumer computed by provider $v1$. Then, provider v1 selects

the consumer with maximum payoff, which is consumer $v7$. Follows, it adds the chosen SMP, meeting point $v2$, to the $ACCEPT$ message (Table 6). Once the $ACCEPT$ message (Table 6) is received by consumer $v7$, it replies back with an ACK message. The recipient of $ACKACK$ message as a reply considers the pair (1,7) is allocated for V2V charging at meeting point $v2$. At consumer $v5$, after τ duration, the $ACCEPT$ (Table 6) Recipient Timer expires while the $ACCEPT$ message (Table 6) is not received. Consequently, it removes provider $v1$ from the Providers Table and repeat the steps of sending $OFFER$ message (Table 5). It sends $OFFER$ message (Table 5) to $v3$, which in return accepts it as it will not receive other offers. Eventually, $v3$ and $v7$ are allocated for V2V charging. Table 14 shows the allocation of consumer-providers charging pairs in the considered scenario.

Table 12: Calculated Payments

	Payment (cents) to	
	Provider v1	**Provider v3**
Consumer v5	100	130
Consumer v7	150	195

Table 13: Payoff of Provider v1

Consumer	**Payoff (cents)**
v5	91.32
v7	128.83

Table 14: Allocated Pairs

Charging Pair ID	**Consumer**	**Provider**
1	v5	v3
2	v7	v1

6. Simulation Results

6.1. Simulation's Parameters and Environment

The proposed protocol is implemented and the simulations are carried out on NS3 [39]. The simulation area is developed using Sumo [40]. The QoS-OLSR [16] is adopted as underlying protocol at the network layer such that the messages, equations, and algorithms proposed in [16] are implemented by modifying the original OLSR on NS3. The simulation area consists of a 10 km highway with 3 lanes. All the lanes have the same movement direction. On the highway side, there are 2 meeting points positioned at 5 km and 10 km respectively. Table 15 shows the simulation parameters; mobility, network, and protocol parameters. The mobility of vehicles is generated using Sumo traces utilizing *Random Trips* and $DUAROUTER$ algorithms to simulate vehicles' real traces [40]. On the highway, vehicles are mobile with a speed range between 60 and 120 km/h. For the network performance results, explained in Section 6.3, the range of the total number of vehicles is between 75 and 300 vehicles generated in the simulation area based on the arrival rate. Whereas to evaluate the charging protocol, 150 vehicles in total with an arrival rate of 1 $vehicle/second$ is considered. At the protocol layer, a varying percentage of providers and consumers are allocated randomly to evaluate the protocol performance. Providers are assigned a selling energy unit price randomly between 10 and 20 $cents/kWh$. While consumers require a charge between 40 and 70 kWh. At the physical layer, IEEE802.11p is adopted with the configuration provided in Table 15. All the results in this section is generated with a confidence level of 95%.

Table 15: Simulation Parameters

Mobility Parameters	**Value**
Number of Vehicles	$[75 - 300]$ $vehicles$
Mobility Model	Sumo Random Trips and DUAROUTER
Arrival Rate	$[0.5 - 2]$ $vehicles/second$
Velocity	$[60 - 120]$ km/h
Network Parameters	**Value**
OLSR Parameters	Default
Routing Bandwidth (BW)	Random percentage of Total Bandwidth
Network Simulator	NS3
HELLO Interval	2 s
TC Interval	5 s
Neighbor Hold Time (Vtime)	6 s ($3 \times HELLO\ Interval$)
Topology Hold Time	15 s ($3 \times TC\ Interval$)
Duplicate Message Hold Time	30 s
Protocol Parameters	**Value**
Providers Percentage	$[10 - 40]\%$
Consumers Percentage	$[10 - 40]$ %
Selling Energy Unit Price (UP_p)	$[10 - 20]$ $cent/kWh$
Original Energy Unit Price (pr_o)	8 $cents/kWh$
Time value (ζ)	10 $cents/h$
Required Energy	$[40 - 70]$ kWh
Charging Efficiency Coefficient (η)	95 %
Charging Rate (ω)	40 kW
Confidence Level	95%
Physical Parameters	**Value**
IEEE 802.11p parameters	Default
Transmission Range	250 m
Total Bandwidth	5 $Mbps$
Data Rate	5 $Mbps$
Transmit Power	33 dBm
Energy Detection Threshold	-61.8 dBm

6.2. Evaluation Metrics

The proposed protocol is analyzed and evaluated based on the network performance and charging pairs allocation performance. The network performance metrics are packet delivery percentage and end-to-end delay. packet delivery percentage is the percentage of the delivered packets to the total packets sent, while end-to-end delay presents the time between sending the packet and receiving it. The allocation performance evaluation and analyzing metrics are:

- **Payoff:** the average payoff computed by providers in *cents*. Each provider payoff is computed considering the profit and total cost of charging. The payoff equation is presented in Equation 3 and explained in Section 3.2.2.
- **Allocation Rate (AR):** the percentage of allocated pairs to the maximum number of allocated pairs. Equation 11 shows the computation of allocation rate.

$$AR = \frac{size(AP)}{Np} \tag{11}$$

where AP is the set of Allocated consumer-provider pairs and Np is the total number of pairs that

can be allocated for charging. Np is computed in Equation 12.

$$Np = minimum(size(P), size(C)) \tag{12}$$

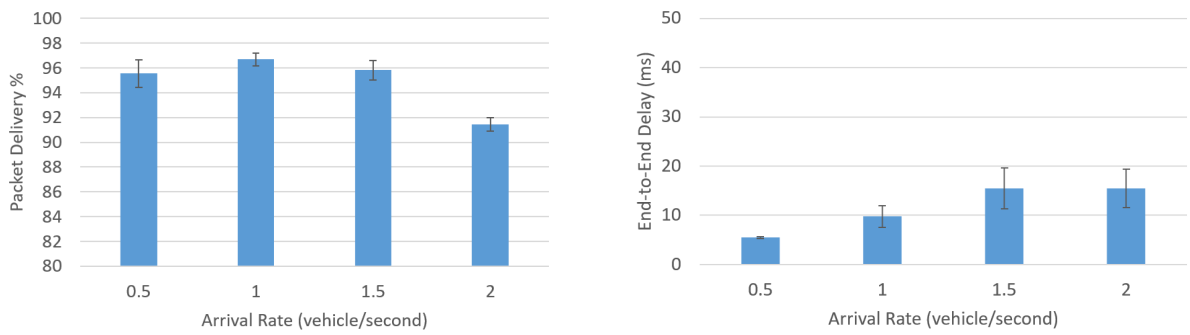
- **Average Reachable Providers:** the average percentage of providers that are reachable by every consumer in the network. A consumer can either reach a provider either through a direct connection within the transmission range or through a propagated PA message (Table 4). In other words, the number of reachable providers for each consumer is the size of its Providers Table.

- **Expected Allocation (EA):** the average allocation rate for the providers-reachable consumers. Providers-reachable consumers have one or more providers in their Providers Table.

The proposed protocol is compared to the centralized allocation approach. We have implemented the centralized allocation such that a central unit, the cloud server, has a global real-time view of all the vehicles in the network. It collects real-time information such as energy selling price of providers, required energy of consumers, preferred meeting point, speed, and location. In contrast to the proposed protocol, vehicles do not share messages among each other to find a a potential energy provider nor complete the V2V charging pair allocation. However, the central unit utilizes the real-time information of all vehicles to allocate providers and consumers. Like the proposed protocol, the charging pair allocation is based on the payoff equation presented in Equation 3. The central unit computes the payoffs of every provider when allocated to every consumer. Consequently, the provider with enough energy to sell with the lowest selling price is allocated to the consumer with the highest payoff. Three scenarios for centralized allocation were simulated and compared to the proposed protocol, namely, centralized allocation with: 1) all vehicles connected (0% disconnected) 2) 20% of providers are disconnected, and 3) 40% of providers are disconnected."

6.3. Network Performance

To show the overhead of exchanging the protocol messages, Figures 3a and 3b present the packet delivery percentage and end-to-end delay results respectively. To evaluate the network performance, the arrival rate of vehicles into the simulation area varies between 0.5 $vehicle/second$ to 2 $vehicle/second$. Accordingly, the packet delivery percentage is maintained above 90% although VANET is characterized by rapid topology change which results in a disconnection between vehicles. Also, the time between sending a message and receiving it at the destination vehicle is recorded in a range between 5 and 15 ms. Hence, the network performance results indicate a reliable delivery of protocol messages with low time delay, resulting in the successful completion of charging pairs allocation with low overhead.



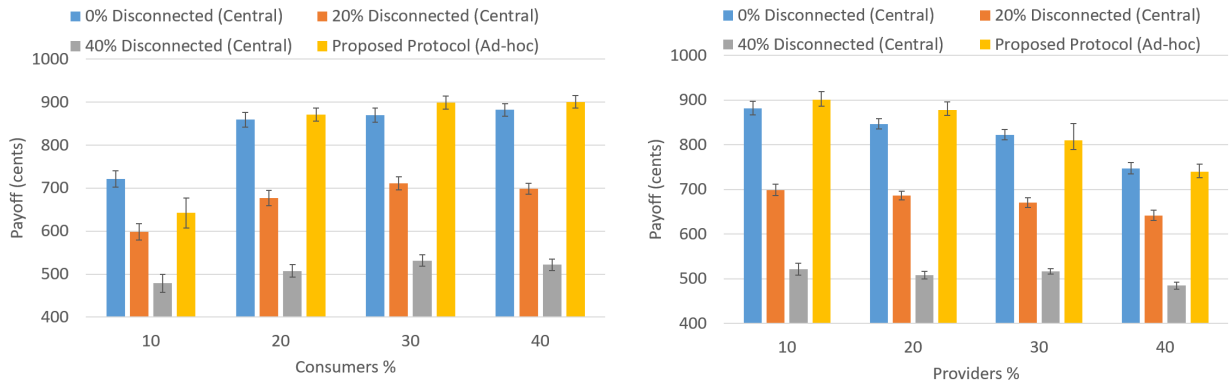
(a) Packet Delivery Ratio

(b) End-to-End Delay

Figure 3: Network performance results considering 20% of the vehicles are consumers and 20% are providers

6.4. Results Analysis and Discussion

In this section, the simulation results of the proposed protocol compared to the centralized allocation, in terms of payoff and allocation rate, are explained. Figure 4a shows the impact of varying the percentage of consumers in the network on average payoff while provider percentage is fixed to 10%. The payoff increases with increasing the percentage of consumers. For centralized allocation, the payoff decreases as the percentage of disconnected providers increases. The proposed protocol payoff performance differs from the centralized allocation by 4.6%. This is a result of the PA message (Table 4) as it allows the periodic discovery of nearby providers. Consequently, the proposed protocol performs better by an average of 16.5% and 52% from centralized allocation with 20% and 40% disconnected providers, respectively.



(a) Payoff with 10% providers

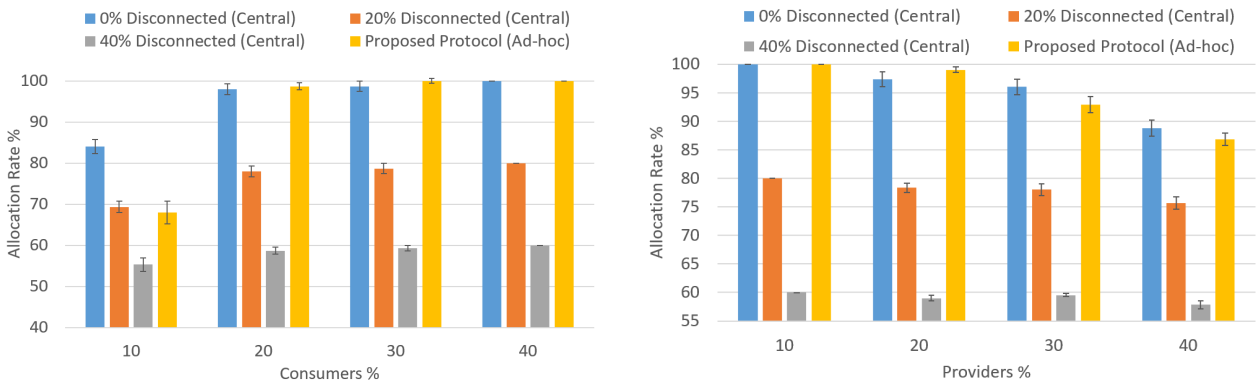
(b) Payoff with 40% consumers

Figure 4: Payoff Results

To evaluate the effect of increasing the percentage of providers from 10% to 40%, the percentage of consumers is fixed to 40% to ensure the allocation of providers. The payoff results of the proposed protocol and centralized solutions are shown in Figure 4b. In the proposed protocol, the consumers send $OFFER$ (Table 5) to the provider with the lowest energy unit price, therefore, the payoff generally decreases as the percentage of providers increases. In comparison with the centralized solution, the proposed protocol has very close payoffs. However, it is outperforming the central unit with 20% and 40% disconnected providers by 9.5% and 42.1% respectively.

With 10% providers and varying percentage of consumers between 10% and 40%, Figure 5a shows the allocation rate of V2V charging pairs. As the providers' percentage is fixed to a lower value than consumers' percentage, it limits the total number of allocated pairs. Therefore, the allocation rate depends on the percentage of connected providers and the energy they are willing to sell. The allocation rate increases as the percentage of consumers increases.The centralized solution with fully connected providers achieves higher than 80% allocation rate and approaches 100% at 40% consumers. However, for the proposed protocol at 10% consumers, performs better than central allocation with 40% disconnected providers. Whereas, it achieves around 100% allocation rate at 30% consumers.

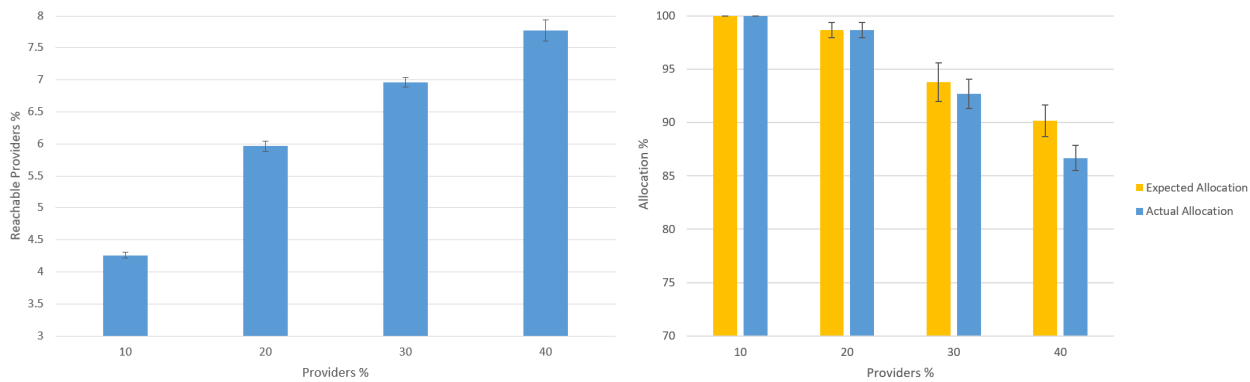
Figure 5b presents the allocation rate of V2V charging pairs considering 40% consumers and varying percentage of providers between 10% and 40%. As consumers' percentage is higher than providers' percentage, the providers determine the percentage of allocated pairs. As the percentage of providers increases, the allocation rate decreases. However, the proposed protocol is allocation rate is maintained above 84%. Although the allocation rate is decreasing as the number of consumers is increasing, the actual allocation rate is away from the expected allocation rate by 1.9% as shown in Figure 6b. The expected allocation rate is based on the percentage of reachable providers by every consumer. The decrease in the allocation rate is due to the average percentage of reachable providers by every consumer, which is limited by the propagation of PA message (Table 4). As shown in Figure 6a, the average percentage of reachable providers increases by 1% when the percentage of providers increases by 10%.



(a) Allocation Rate with 10% providers

(b) Allocation Rate with 40% consumers

Figure 5: Allocation Rate Results



(a) Percentage of Reachable Providers

(b) Expected Allocation Rate

Figure 6: Analysis results of the proposed protocol

7. Conclusion

This paper presents an efficient solution for V2V charging pairs allocation in a decentralized and ad-hoc manner. The protocol is developed at the application layer on top of QoS-OLSR. The QoS-OLSR main task is to compute routing tables performed by MPRs. MPRs are utilized by the proposed protocol to propagate the information of the providers to all vehicles through PA messages (Table 4). Accordingly, the consumers propose to the provider with the least selling price by sending an $OFFER$ message (Table 5). In response, the providers that receive offers reply by an $ACCEPT$ message (Table 6) to the proposed consumer with the highest payoff. Simulations were conducted using NS3 where the simulation area is a highway generated with vehicles traces using Sumo. The results show that the proposed protocol messages are delivered reliably with high packet delivery percentage above 90% and low end-to-end delay recorded in microseconds. On the other hand, the proposed protocol is outperforming the central allocation when 20% and 40% of providers are disconnected in terms of allocation rate and payoff. While the proposed protocol performance is close to the fully connected central unit by 4.6% in terms of payoff and 0.8% in terms of allocation rate.

8. Acknowledgment

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